Database and Data Governance

Relational Database Service (RDS)

RDS for MySQL

RDS for PostgreSQL

RDS for MariaDB

- A service that manages database engines and resources
- Compatible with mainstream relational database engines: **MySQL, PostgreSQL, MariaDB**
- A **cloud-native** service that **facilitates maintenance**
- Built-in **automated backup and patching**
- HA configuration and **automatic switchover/failover** within seconds
- Read replicas
- Security certifications

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Relational Database Service

Relational Database Service (**RDS**) provides a comprehensive performance monitoring system, multi-level security protection measures, and a professional database management platform, allowing you to easily set up and scale a relational database. The console simplifies operation procedures and reduces routine O&M workloads, so that you can focus on your application and service development.

On-premises Databases

- Hardware and software (including servers, systems, and databases) purchase and installation required
- Expensive equipment room hosting
- High DBA costs

VS

Self-Built Databases on ECSs

- Software purchase and installation required
- Cloud server rent fees
- High DBA costs

VS

RDS (MySQL, PostgreSQL, SQL Server)

- **No need to purchase and install any hardware or software**
- **Only RDS rent fees**
- **Low DBA costs**

On-premises Databases	Self-Built Databases on ECSs	RDS (MySQL, PostgreSQL, SQL Server)
Application optimization	Application optimization	Application optimization
Database elastic scaling	Database elastic scaling	Database elastic scaling
Database high availability	Database high availability	Database high availability
Data backup and restoration	Data backup and restoration	Data backup and restoration
Database software upgrade or patch installation	Database software upgrade or patch installation	Database software upgrade or patch installation
Database software installation	Database software installation	Database software installation
OS version upgrade or patch installation	OS version upgrade or patch installation	OS version upgrade or patch installation
OS installation	OS installation	OS installation
Server deployment and maintenance	Server deployment and maintenance	Server deployment and maintenance
Rack stacking	Rack stacking	Rack stacking
Infrastructure (equipment rooms, power supplies, air conditioners, and networks)	Infrastructure (equipment rooms, power supplies, air conditioners, and networks)	Infrastructure (equipment rooms, power supplies, air conditioners, and networks)

Solved by yourself Solved by cloud service providers

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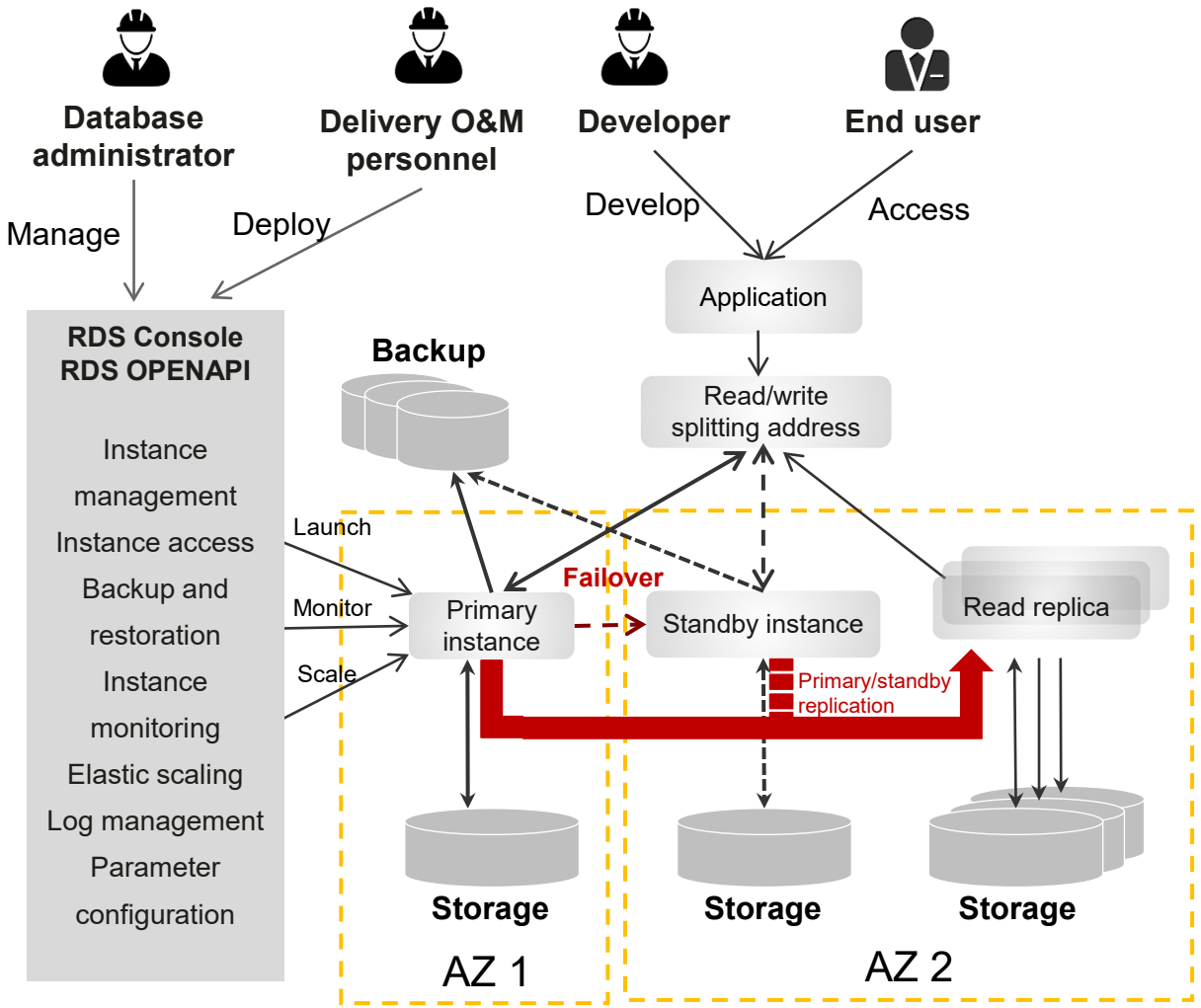
Primary/Standby Deployment of RDS



- If the primary DB instance fails, workloads can be automatically switched to the standby DB instance.
  - The switchover duration varies depending on your DB engine. A switchover can be performed within seconds.
- Primary and standby DB instances can be deployed in different AZs to enable cross-AZ disaster recovery (DR).
- You can complete maintenance operations for the standby database and then perform a switchover to ensure service continuity.
- Primary/standby deployment increases costs.

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Backup Solution

Full backup: A full backup is to back up all data, even if no data has changed since the last backup.Full backups include automated backups and manual backups.

Incremental backup (**MySQL binlog** backup/**PostgreSQL Write-Ahead Logging (WAL)**): RDS automatically backs up data modifications made after the most recent full or incremental backup **every five minutes.**

Automated backups : RDS automatically creates full backups for your instance **during a backup window** you specified and saves the backups based on the configured retention period. If necessary, you can **restore data to any point in time** within the backup retention period.

- **Automated backups cannot be manually deleted.** To delete them, you can adjust the retention period specified in your automated backup policy. When the retention period expires, automated backups will be deleted or remove instances

Manual backups: are user-initiated full backups of instances. The backup method is physical backup.

- **Manual backups will not be deleted until you delete them manually**.

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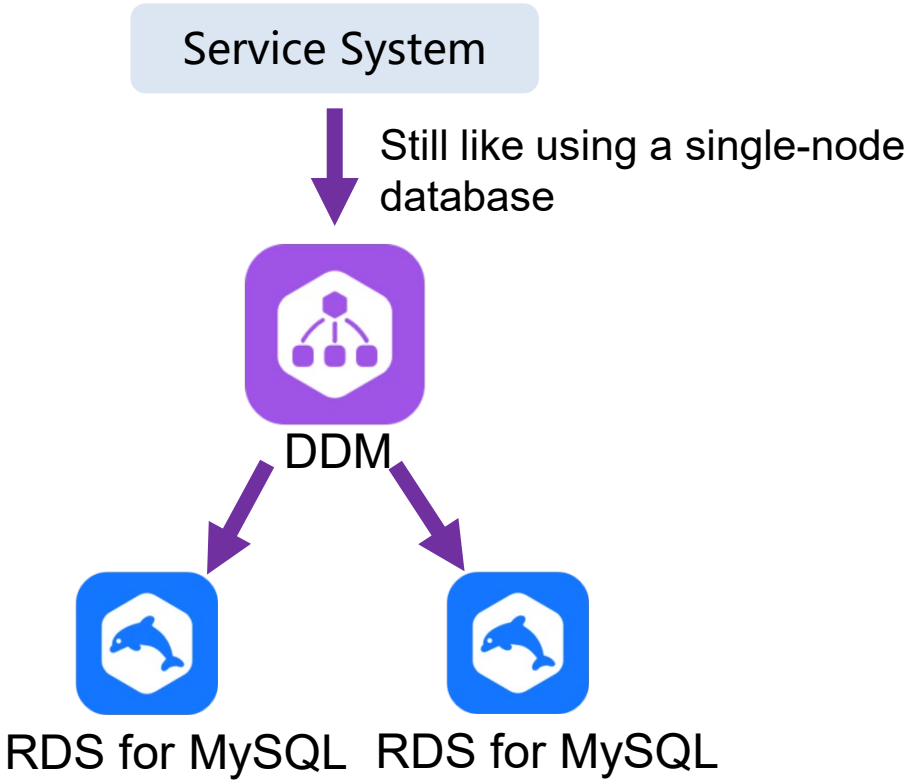
Distributed Database Middleware (DDM)

Distributed Database Middleware (DDM) is a distributed database middleware service that can work with RDS to **horizontally expand database capacity** so that regular users can use distributed databases as easily as using a single-node database.



- SQL computing and transaction processing
- Horizontal database scaling
- Unlimited scaling of storage and compute resources
- Single-node MySQL database experience

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GaussDB (for MySQL)

GaussDB for MySQL

- Huawei DB engine
- Compatible with MySQL
- Several times higher performance than that of native MySQL databases
- Strong backup and restoration capabilities achieved with decoupled storage and com
- High availability in cross-AZ deployment

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GaussDB (for Mongo)

GaussDB for Mongo

Document Database Service (Mongo)

- Huawei-developed DB engine
- Compatible with MongoDB clients
- Compute resources scaled within minutes and storage within seconds
- Cross-AZ HA
- GaussDB(for Mongo) clusters provide high availability

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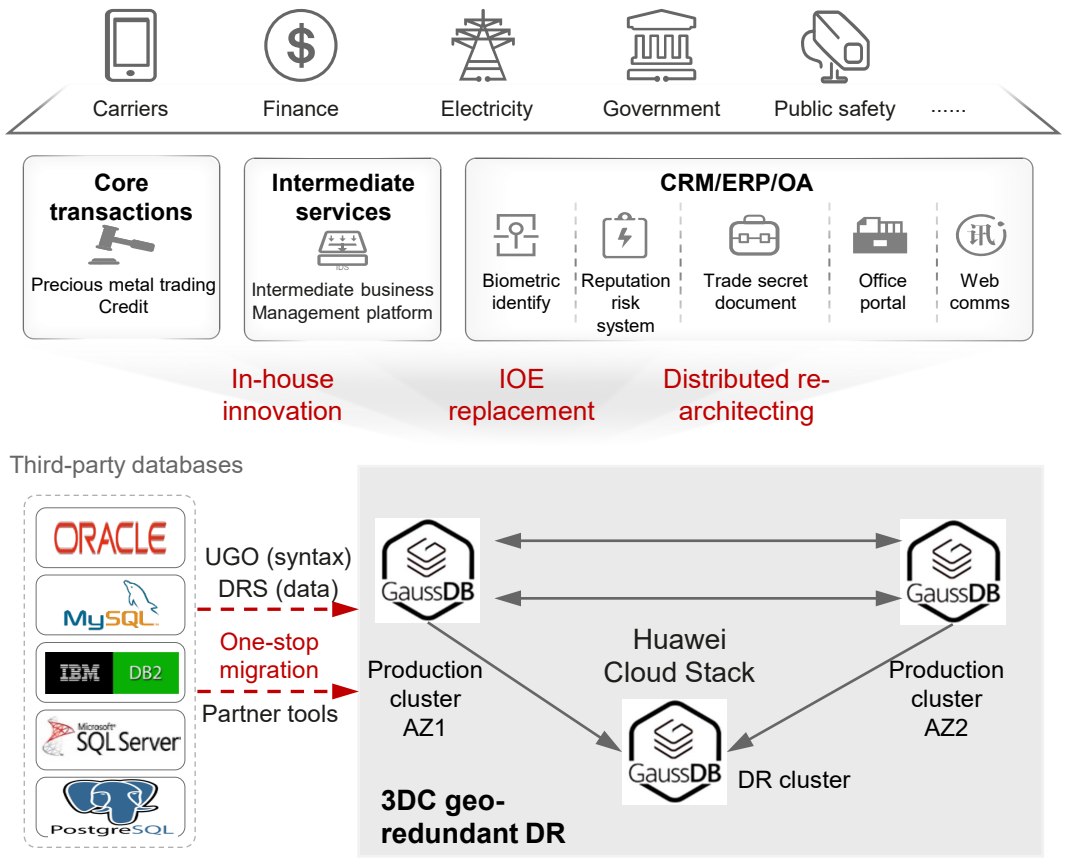
GaussDB: A Flagship Transactional Database from Huawei

GaussDB is fully developed by Huawei based on a full-stack open ecosystem.

Scenarios



Advantages

Availability: Intra-city active -active and remote DR solutions; the first database in China to realize zero RPO in intra-city dual-cluster DR.
Intra-city DR: RPO = 0, RTO < 60s. Remote DR: RPO < 10s, RTO < 10 min

Performance: The only database in China to use the in-place update storage engine, Ustore, to ensure 24/7 high-performance, low-latency processing in all scenarios
Single-node instance: 1.5 million tpmC; 32-node distributed instance: 15 million tpmC

Security: The only database in China to have earned CC EAL4+ certification (source code is highly secure)
Dynamic anonymization, unified audit, and full encryption, and anti-tamper functions, and

Flexibility: Cloud native distributed architecture; flexibly online scale-out within seconds
Large-scale distributed clusters with 1,000+ nodes

Intelligence: The first AI-native database in China
Intelligent recommendation for indexes and distribution columns helps a large bank to improve the performance by 15%. Intelligent fault root cause analysis improves the fault diagnosis by 5x.

Migration: Highly compatible with Oracle; DRS+UGO one-stop migration
Semantic-aware data compression reduces storage costs by 50%. Kernel multi-tenant mode improves resource utilization by 4x.

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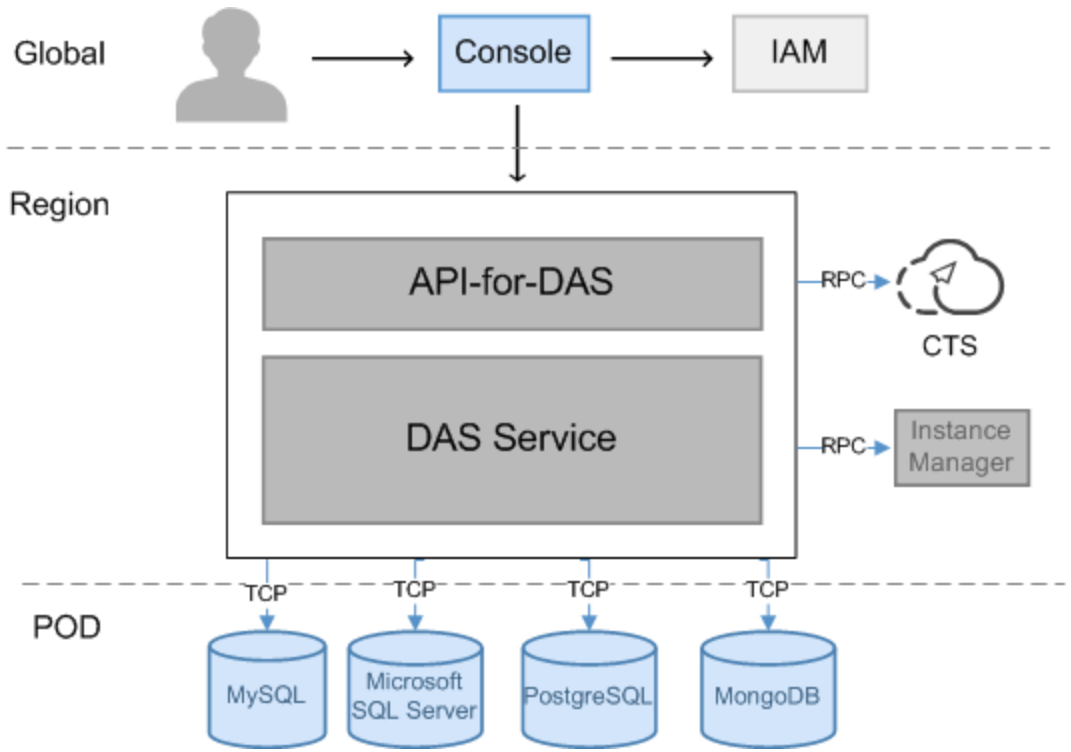
Data Admin Service (DAS)

Data Admin Service (DAS) enables you to manage DB instances on a web-based console, simplifying database management and improving working efficiency.



SQL statement execution

- SQL formatting
- SQL intelligent prompt
- SQL plan execution
- SQL template

Data query and editing

- Online query and data editing
- Data import and export
- INSERT statement and CSV format

Management of databases, tables, and views

- Creating and altering objects
- Exporting table structure DDL
- Comparing table structure DDL

User and object rights

- Permission to connect to databases
- Permission to access objects
- Global permission

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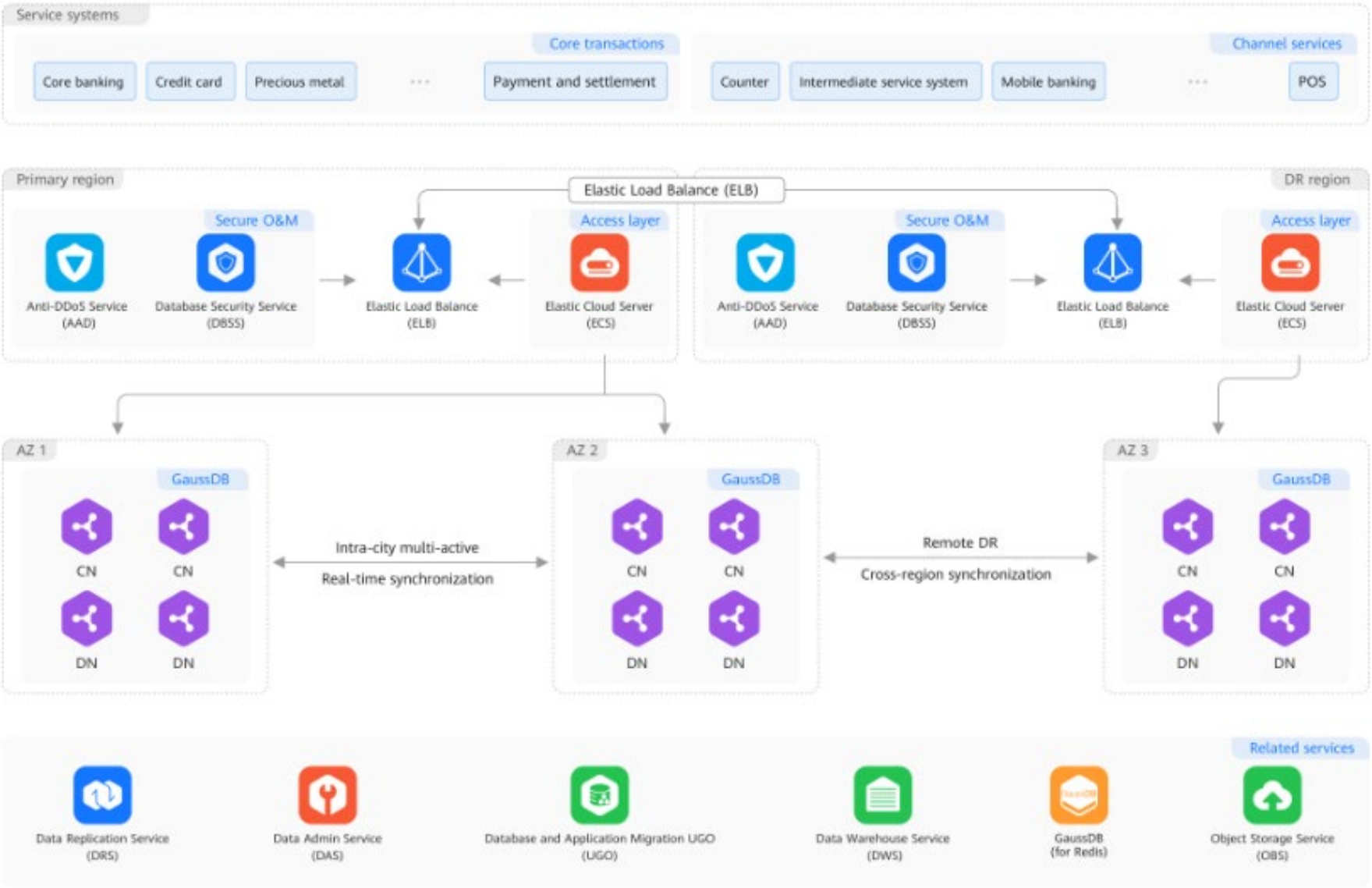
Service systems
Core transactions
Core banking
Credit card
Precious metal
...
Payment and settlement
Channel services
Counter
Intermediate service system
Mobile banking
...
POS
Primary region
Elastic Load Balance (ELB)
DR region
Secure O&M
Anti-DDoS Service (AAD)
Database Security Service (DBSS)
Elastic Load Balance (ELB)
Access layer
Elastic Cloud Server (ECS)
Secure O&M
Anti-DDoS Service (AAD)
Database Security Service (DBSS)
Elastic Load Balance (ELB)
Access layer
Elastic Cloud Server (ECS)
AZ 1
GaussDB
CN
CN
DN
DN
Intra-city multi-active
Real-time synchronization
AZ 2
GaussDB
CN
CN
DN
DN
Remote DR
Cross-region synchronization
AZ 3
GaussDB
CN
CN
DN
DN
Related services
Data Replication Service (DRS)
Data Admin Service (DAS)
Database and Application Migration UGO (UGO)
Data Warehouse Service (DWS)
GaussDB (for Redis)
Object Storage Service (OBS)

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DataArts Studio

DataArts Studio

- Starts from data governance planning.
- Data import
- One platform for data development and governance
- Data quality evaluation based on planning
- Data asset management and metadata splitting

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Data Collection

Transaction data
Structured transaction information

RDS for MySQL

RDS for PostgreSQL

File Log, and raw data

OBS

Stream data, IoT, and Clickstreams

DMS for Kafka

Data Ingestion service

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Huawei Lakehouse Architecture Design Framework

Data Sources

- Data Stream
- RDMBS
- Rest-APIs
- Batch Files

Data Ingestion

- **DMS-Kafka** – Stream Processing
- **Database Replication Service** – Database/Table Sync
- **IOTDA** – IOT Data Sync
- **Object Migration Service** – Object Storage Sync
- **Cloud Data Migration** – Batch Data Ingestion

****services to be provisioned as needed****

Data Storage

Object Storage Services

- Raw Data
- Transformed Data
- Enriched Data

CSV, JSON, Parquet, Apache orc, AVRO

Data Analytics

Data Warehouse Service

On-Demand Virtual Warehouses

Data Lake Insight or MapReduce Service

hadoop, Spark, Flink, trino

On-demand Analytics Engine

Data Application

ModelArts

AI/ML Development

Data Analytics

Spark, python, PySpark, Scala, jupyter, Java

Dashboard / Self Service Report

tableau, Power BI

Mobile Apps/ Website/ …

Application can be deployed on HWC, on-prem, or other CSPs

DataArts Studio: DataArts Migration | DataArts Factory | DataArts Catalog | DataArts Quality | DataArts DataService

Heterogenous data sources on premise and on-cloud	Range of services to ingest data from different type of sources into Data Lake	High Performance Object Storage, fully optimized for lake house workload	Scalable on-demand compute resources to query data from Object storage	Industry Standard Connector (Rest-API, PostgreSQL, S3, HDFS API, JDBC/ODBC)

Huawei Public Cloud / Huawei Cloud Stack

Map Reduce Service (MRS)

MapReduce

- Provided open-source Hadoop-based components
- Deploy a Hadoop cluster with a few clicks.
- Easily run big data components such as Storm, Hadoop, Spark, HBase, Kafka and Flink
- fully compatible with open source APIs and Huawei Cloud computing and storage
- Technical support provides multiple support services
    - Creating, deleting, and scaling in or out a cluster
    - Managing cluster jobs
    - Managing cluster alarms
    - Managing cluster patches
    - Managing IAM users
    - Managing external APIs

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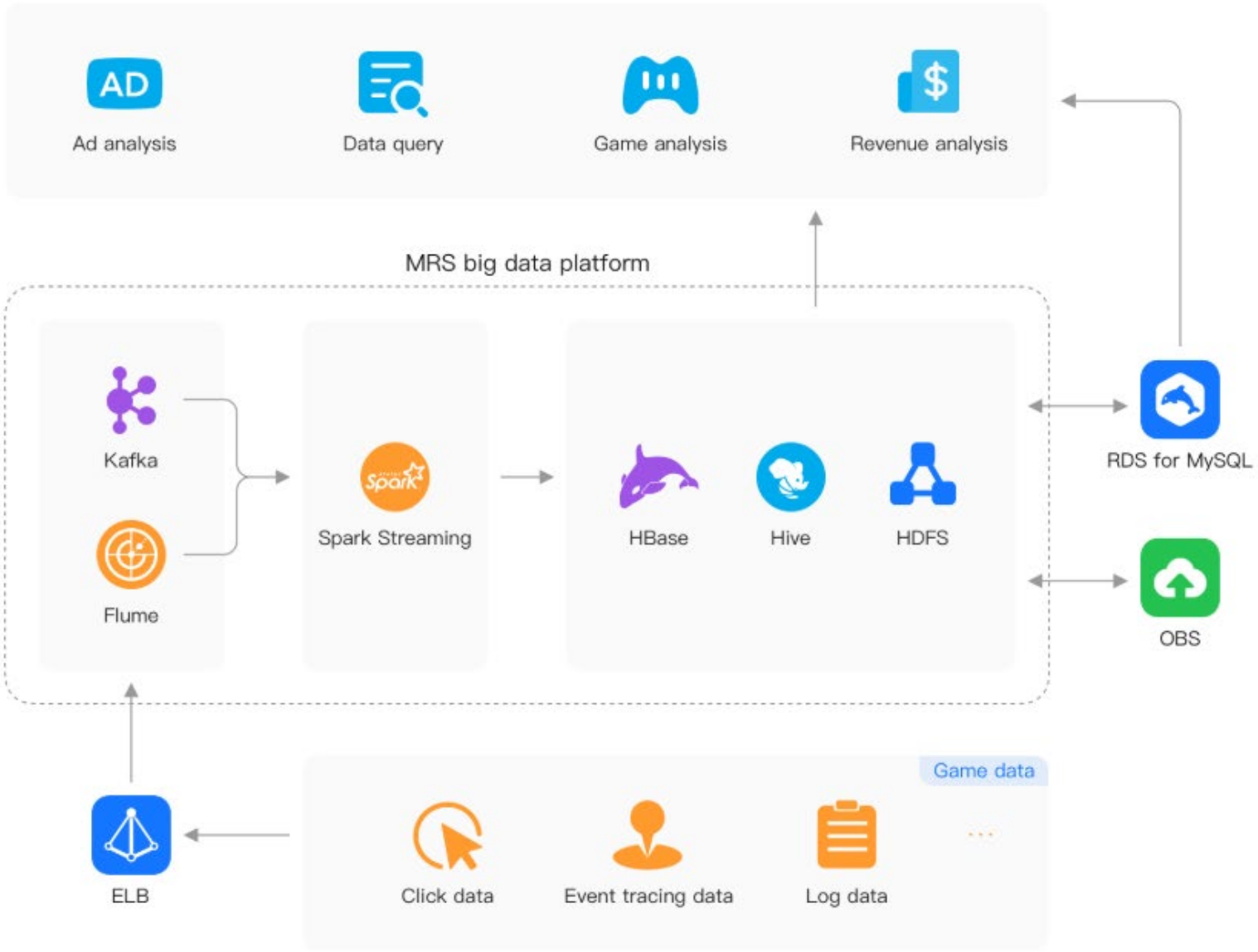
AD
Ad analysis
Data query
Game analysis
Revenue analysis
MRS big data platform
Kafka
Flume
Spark Streaming
HBase
Hive
HDFS
RDS for MySQL
OBS
ELB
Game data
Click data
Event tracing data
Log data
...

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Cloud Search Service (CSS)

Cloud Search Service (CSS)

- Compatible with open-source Elasticsearch, Kibana, and Cerebro
- For structured and unstructured data search, and use AI vectors for combine search, statistics, and reports
- Store the snapshot data of Elasticsearch instances to OBS
- Example use cases:
  - Site search (Full-text search)
  - Log analysis
  - Vector search (unstructured data, such as images, videos, and corpuses)

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